

S12 – Combined Sensitivity Analysis

Purpose

The purpose of this analysis is to integrate all previously evaluated uncertainty dimensions — including clause mapping variability, AHP weight perturbation, seed-level model variability, and edge-feasibility thresholds — into a unified robustness assessment. The objective is to determine whether any plausible combination of uncertainties could overturn the study's principal governance recommendation.

Methodology

A pessimistic worst-case sensitivity configuration was constructed using the following combined perturbations:

Regulatory Clause Mapping

All mapped clauses were reassigned to the next more adverse regulatory embedding level, consistent with the sensitivity procedure described in Section S8.

AHP Weight Perturbation

AHP weights were shifted by +0.03 toward dimensions with lower baseline scores (Accuracy and F1 Score) and away from Compliance and Edge Readiness dimensions.

Seed-Level Model Variability

The lowest Clinical Deployment Index (CDI) value observed across the ten experimental seeds for HIMS-CDI_cal was selected from the bootstrap distribution reported in Table 5.

Edge Threshold Constraints

The strictest deployment constraints examined in Section S11 were applied:

- Latency ≤ 100 ms
- Memory ≤ 50 MB

The resulting configuration, therefore, represents the most adverse plausible governance scenario considered in the study.

Results

Extreme Threshold Scenario

Under the strictest edge thresholds (100 ms latency and 50 MB memory):

- No evaluated model satisfies both deployment constraints simultaneously.
- HIMS-CDI_cal exceeds the latency threshold by a slight margin (132 ms > 100 ms).

However, these constraints are substantially more restrictive than currently proposed clinical edge requirements and are therefore not considered operationally realistic.

Governance-Gate Stability

Across all other perturbation combinations:

- HIMS-CDI_cal remained the only model satisfying:
 - the CDI governance threshold ($CDI \geq 0.70$),
 - the Clause Coverage Rate threshold (≥ 0.80), and
 - realistic edge-feasibility constraints simultaneously.

Additionally:

- Weight perturbations and clause-mapping shifts reduced CDI values by no more than 0.02.
- The minimum observed CDI remained above 0.75.
- No perturbation configuration caused HIMS-CDI_cal to fail either:
 - the CDI governance gate, or
 - the Clause Coverage Rate requirement.

Recommendation Stability

The analysis demonstrates that the recommendation of HIMS-CDI MLR as the preferred edge-ready governance architecture remains invariant under all plausible perturbation scenarios considered within the synthetic evaluation framework.

Importantly:

- No alternative model simultaneously achieved:
 - acceptable governance compliance,
 - operational edge feasibility, and
 - stable deployment readiness.

Therefore, the combined sensitivity analysis confirms that the governance-first deployment recommendation remains internally consistent and robust to uncertainty across all evaluated dimensions.

Conclusion

The combined sensitivity analysis demonstrates that the proposed governance-gate architecture remains robust under all tested uncertainty conditions. Under realistic deployment thresholds, HIMS-CDI_cal remains the only model that simultaneously satisfies all governance requirements.

Neither individual perturbation nor any combined-uncertainty scenario reverses this outcome. Consequently, the provisional deployment criteria, governance thresholds, and edge-readiness recommendations remain internally coherent and operationally stable within the synthetic evaluation framework presented in the manuscript.